

--	--	--	--	--	--	--	--

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2024

ME 19-205-0705 SUPPLY CHAIN MANAGEMENT (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Define a supply chain and understand its different structures and its importance to the success of a firm.
- CO2: Understand the concept of strategic fit between the SC strategy and the competitive strategy of the firm and how to achieve it.
- CO3: Identify the main drivers of supply chain (SC) performance and measure the musing precise metrics.
- CO4: Assess the importance of distribution networks and the different opinions available for their design.
- CO5: Develop a methodological framework for network design and facility location and use optimisation models and techniques for facility location and capacity allocation.
- CO6: Use safety inventory to deal with demand uncertainty in the supply chain.
- CO7: Understand the importance of transportation in the SC, identify the different modes of transportation and the role played by the infrastructure and policies.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

(8 × 3 = 24) Marks BL CO PO

- | | | | | | |
|--------|--|---|----|---|---|
| I. (a) | How does supply chain management contribute to cost reduction and efficiency improvement in a company? | 3 | L1 | 1 | 1 |
| (b) | What is the significance of supply chain integration and how can companies achieve it? | 3 | L2 | 2 | 1 |
| (c) | Explain the concept of Economic Order Quantity in inventory management in supply chain. | 3 | L2 | 3 | 1 |
| (d) | What is safety stock and how it is calculated? | 3 | L3 | 6 | 1 |
| (e) | How do transportation costs impacts supply chain decisions and overall product pricing? | 3 | L3 | 4 | 2 |
| (f) | What are the key challenges in managing transportation within global supply chains? | 3 | L2 | 7 | 1 |
| (g) | How does the proximity to suppliers and customers affect facility location decisions in supply chain management? | 3 | L3 | 5 | 1 |
| (h) | How can companies optimize capacity allocation to minimize costs while meeting customer demand? | 3 | L2 | 5 | 1 |

PART B

(4 × 12 = 48)

- | | | | | | |
|---------|---|---|----|---|---|
| II. (a) | What are the key supply chain decision phases? Explain. | 4 | L2 | 1 | 1 |
| (b) | Distinguish between the push and pull views of a supply chain. | 4 | L1 | 1 | 1 |
| (c) | Explain why achieving strategic fit is critical to a company's overall success. | 4 | L3 | 2 | 2 |

OR

- | | | | | | |
|----------|--|---|----|---|---|
| III. (a) | What are the key components or stages of a typical supply chain? | 4 | L2 | 1 | 1 |
| (b) | Explain the role of logistics in supply chain management. | 4 | L2 | 4 | 2 |
| (c) | How does a supply chain function within a business? | 4 | L2 | 3 | 2 |

(P.T.O.)

BTS-VII(R/S)-11-24-2974

		Marks	BL	CO	PO
IV.	(a) What are the main types of inventory and why is managing each type important in supply chain operations?	3	L2	5	1
	(b) How does inventory turnover ratio help in assessing the efficiency of inventory management?	4	L2	5	2
	(c) A retailer sells 1000 units of a product weekly. The lead time to restock is 4 weeks and the standard deviation of weekly demand is 50 units. The desired service level is 95% and the Z-value for 95% is 1.65. Calculate the required safety stock.	5	L3	6	2
OR					
V.	(a) A company has an average daily demand of 80 units. The lead time for orders varies with an average of 10 days and a standard deviation of 2 days. The company wants 99% service level with a Z-value of 2.33. What is the reorder point (ROP)?	7	L3	6	2
	(b) What is the impact of inventory centralization on various supply chain performance measures?	5	L2	5	2
VI.	(a) What are the main drivers of trucking rate structures?	5	L2	4	1
	(b) A firm wants to design incentive systems for its transport contractor so as to improve transport lead-time reliability. To design this system they want to quantify the benefits of transport lead time reliability. Suggest a suitable approach for this quantification exercise.	7	L2	7	1
OR					
VII.	(a) What are the benefits of cross-docking? What are the difficulties in implementing cross-docking?	6	L3	5	2
	(b) Explain the concept of inter-modal transportation and its benefits in modern supply chain operations.	6	L3	5	2
VIII.	(a) What is the significance of cost considerations (labour, transportation, taxes etc.) in facility location selection?	6	L3	5	4
	(b) What role do Geographic Information Systems (GIS) play in optimizing facility location decisions?	6	L3	5	4
OR					
IX.	(a) How does demand variability affect capacity allocation decisions and what strategies can be used to address it?	6	L3	5	4
	(b) What strategies can be implemented to reduce or eliminate the bull whip effect in supply chain operations?	6	L3	5	4

Bloom's Taxonomy Level's

L1 – 9.17%, L2 – 38.33%, L3 – 52.50%.

